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Battery Storage Market Notes, August 2026 issue

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"Grid-scale battery storage" means large batteries connected to the power grid that store electricity and release it. They earn money by selling the "capability" to fix electricity surpluses and shortages. They also earn from the way the price of electricity changes by time of day. These notes use published figures and our own records to show this month's movements in charts and tables. This issue is a back issue, made in November 2026 for an earlier month. The grid connection study figures are as of when the issue was made (October 19, 2026).

1. Three income streams for a battery storage plant (how to read these notes)

A battery storage plant (grid-scale battery) earns income from three markets. ① In the balancing market (EPRX), it sells the "capability" to fix electricity surpluses and shortages to grid operators. ② In wholesale electricity (JEPX), it buys and stores electricity in cheap hours and sells it in expensive hours. ③ In the capacity market (OCCTO), the country buys future electricity supply capacity in advance. These notes list the figures for these three markets in this order. The ①, ② or ③ at the start of each section title shows which market the section is about.

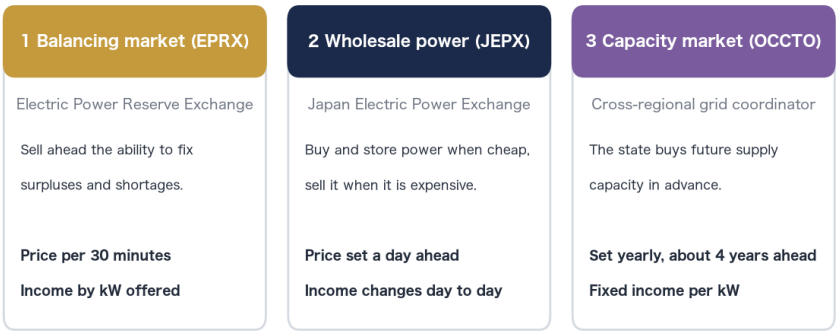


Figure 1 Three income streams for a battery storage plant

Source: Market rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three trends this month

We picked three movements this month from the figures of the three markets.

- In the balancing market (EPRX), the bid ratio for Primary reserve (online) went above 100% in June 2026, at 106.8%. This month it was 116.0%, and the average winning price was 4.40 yen/kW per 30 min.
- In the balancing market (EPRX), 52.2% of the August 2026 sales of Tertiary reserve 2 came from the top 10% of 30-minute slots by sales.
- In wholesale electricity (JEPX), we compared the average price of the 4 most expensive hours with the average of the 4 cheapest hours. In August 2026, the monthly average of this gap was 11.94 yen/kWh in Tokyo and 18.82 yen/kWh in Kyushu.

3. ① Balancing market (EPRX) | 7 products, and prices by area

The balancing market is where grid operators buy, every 30 minutes, the "capability" to fix electricity surpluses and shortages. Products are split by how fast they change output. Primary reserve responds to changes in frequency. Secondary and Tertiary reserves respond to instructions from the grid operator. Primary reserve is the fastest, and Tertiary reserve 2 is the slowest. Primary reserve senses frequency changes by itself and responds. It has two categories. In "online", the grid operator watches its operation over a dedicated line. In "offline", no dedicated line is laid, and the record of operation is submitted afterwards. They are sought and traded separately, so we show them separately. A Composite product is a bid that offers several products together. Prices are set in yen per 1 kW of capability per 30 minutes. kW is a unit of how much power can be delivered. It is different from kWh, which measures an amount of electricity. Each 30-minute trading round is called a "30-minute slot". The figure shows the range of winning prices by area (by area where the power source is). The thin line runs from the cheapest bid to the most expensive bid in the month. The thick band is the price range that holds 8 out of every 10 slots with winning bids. The dot is the average winning price. Nationwide figures are in the table in the "Market figures" section.

By area where the power source is, the highest average winning price in the balancing market was 14.52 yen/kW per 30 min for Primary reserve (offline) in Tohoku. The lowest was 0.38 yen/kW per 30 min for Tertiary reserve 2 in Shikoku.



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Figure 2 Range and average of winning prices by area in the balancing market (August 2026 (1–31), by area where the power source is)

Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

Product	Response time	Duration
Primary reserve (online)	Within 10 seconds	5 minutes or more
Primary reserve (offline)	Within 30 seconds	No set duration
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes
Composite product	Requirements of the combined products	Requirements of the combined products

4. ① Balancing market (EPRX) | Grid areas and times where bids win more easily

The clearing rate is the share of the amount bid that won. The higher it is, the more easily bids win. The figure shows the clearing rate for each product by time of day (every 30 minutes), for each grid area (area where the power source is). Where a line breaks, there were no bids at that time.

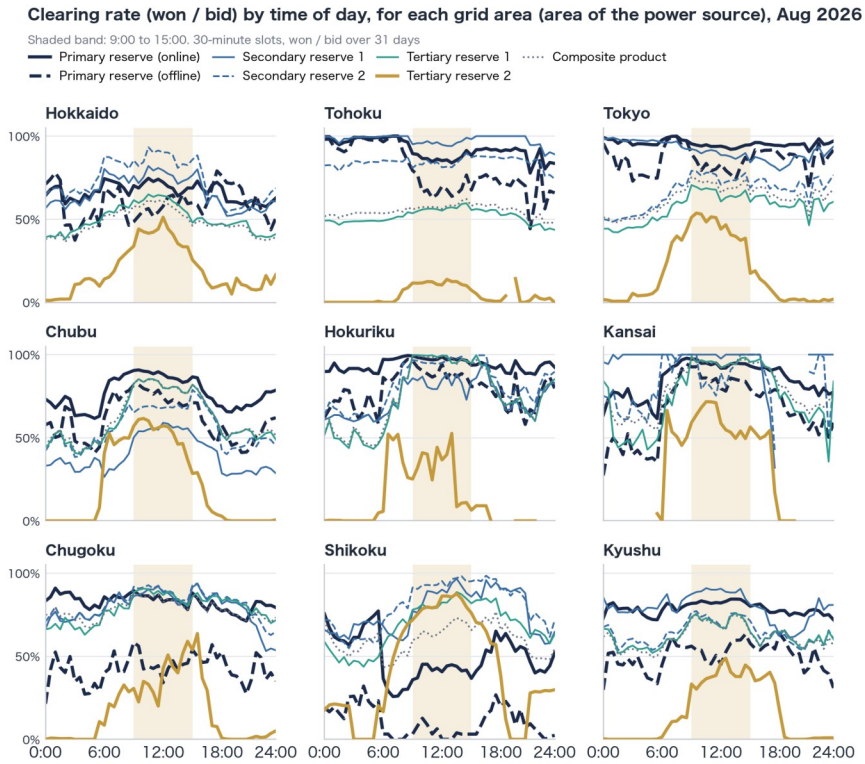


Figure 3 Clearing rate in the balancing market (grid area (area where the power source is) × time of day, August 2026 (1–31))

Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

5. ① Balancing market (EPRX) | Primary reserve (online) became a fight for seats

Primary reserve is the fastest product. It responds to changes in frequency and changes output within 10 seconds (online) or within 30 seconds (offline). Batteries are often used for it. The left side is the online category, and the right side is the offline category. The bars show the bid ratio (amount bid ÷ amount sought by grid operators). Above 100% means more bids came in than were sought. Even below 100%, some bids do not win (see the clearing rate in the "Market figures" table). The bars rise when bids increase, and also when the amount sought falls. The line is the average winning price. These are monthly figures from April 2025, the data we have.

The bid ratio for Primary reserve (online) in the balancing market went above 100% in June 2026, at 106.8%, for the first time since April 2025. This month it was 116.0% (amount sought: 3,066.1 MW in April 2025, 1,507.8 MW this month). The average winning price went from 2.89 yen/kW per 30 min in April 2025 to 4.40 yen/kW per 30 min this month. For Primary reserve (offline), the bid ratio this month was 73.4% (amount sought: 1,122.5 MW in April 2025, 497.4 MW this month), and the average winning price was 9.63 yen/kW per 30 min.

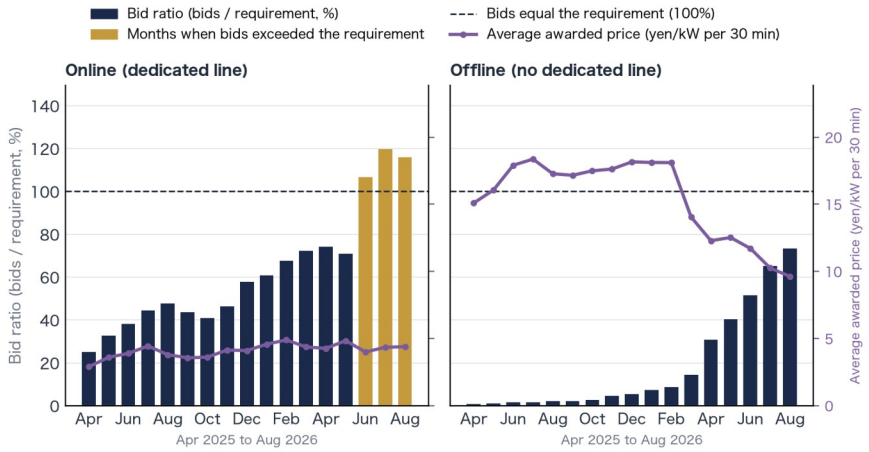


Figure 4 Bid ratio (bars) and average winning price (line) for Primary reserve (monthly; left is online, right is offline; gold bars are months when bids exceeded the amount sought)

Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

6. ① Balancing market (EPRX) | How prices by product have moved

These are average winning prices by product since April 2025 (monthly averages weighted by the amount won). From March 14, 2026, Primary reserve to Tertiary reserve 1 and Composite products moved from trading every 3 hours to trading every 30 minutes (Tertiary reserve 2 was already traded every 30 minutes).

Average winning prices in the balancing market (all in yen/kW per 30 min) moved from April 2025 to this month as follows: Primary reserve (online) 2.89 yen → 4.40 yen, Primary reserve (offline) 15.10 yen → 9.63 yen, Tertiary reserve 2 1.12 yen → 2.03 yen. Their highest months were, in the same order, February 2026 at 4.90 yen, July 2025 at 18.37 yen and May 2026 at 4.18 yen. For Secondary reserve 1, Secondary reserve 2, Tertiary reserve 1 and Composite products, the highest month was July 2025.

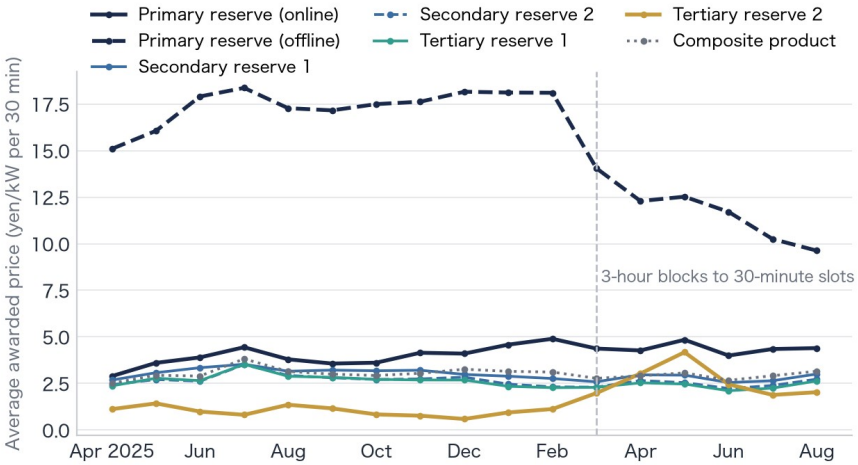


Figure 5 Average winning price by product in the balancing market (yen/kW per 30 min, monthly)
Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

7. ① Balancing market (EPRX) | Which hours of the day are expensive

These are average winning prices by product for each hour, August 2026 (1–31).

For Primary reserve (online) in the balancing market, the average winning price was highest in the 18:00 hour at 5.53 yen/kW per 30 min, and lowest in the 0:00 hour at 3.50 yen/kW per 30 min.

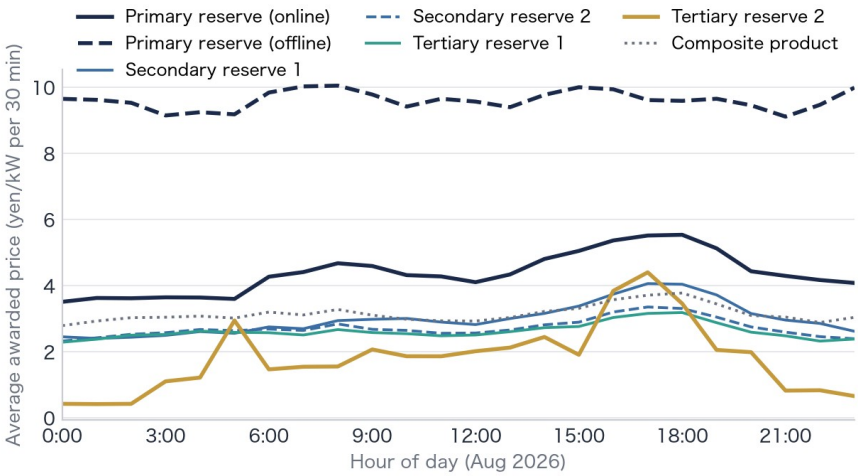


Figure 6 Average winning price by time of day in the balancing market (August 2026 (1–31), yen/kW per 30 min)
Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

8. ① Balancing market (EPRX) | Which products batteries are used in

The figure shows the share of the amount won that batteries account for. The table compares the average winning price of the amount batteries won with the average winning price of the whole product.

In Primary reserve (online) in the balancing market, batteries had a 24.4% share. The average winning price of the amount batteries won was 8.08 yen/kW per 30 min (4.40 yen/kW per 30 min overall).

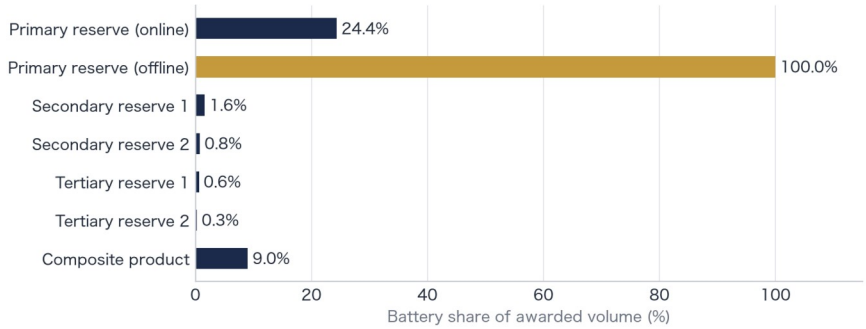


Figure 7 Share of batteries in the balancing market (%)

Source: Trading results (preliminary figures) by power source type, published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

Product	Won by batteries (yen/kW per 30 min)	Whole product (yen/kW per 30 min)
Primary reserve (online)	8.08	4.40
Primary reserve (offline)	9.63	9.63
Secondary reserve 1	5.14	3.01
Secondary reserve 2	4.08	2.73
Tertiary reserve 1	4.03	2.63
Tertiary reserve 2	51.66	2.03
Composite product	8.02	3.15

Both are averages weighted by the amount won (August 2026, preliminary figures, nationwide). In Tertiary reserve 2, batteries account for only 0.3% of the amount won. In the slots that batteries won, the nationwide average winning price, weighted by the amount batteries won, is 6.54 yen. The battery average is high not because batteries lean toward high-priced slots. It is high because the batteries' bid prices themselves are high (each winning bid is settled at its own bid price).

9. ① Balancing market (EPRX) | Sales concentrate in a small number of slots

We ranked the month's 30-minute slots with winning bids by sales (average winning price × amount won, nationwide), from highest to lowest. For each product, we plotted what % of the month's sales the top slots account for. The more a curve bulges above the dotted line, the more sales are concentrated in fewer slots. Tertiary reserve 2 is a product that covers errors in forecasting renewable output caused by the weather. Its price is low in most slots and high in a very small number of slots.

The share of balancing market sales concentrated in the top 10% of slots was 52.2% for Tertiary reserve 2 and 15.5% for Primary reserve (online).

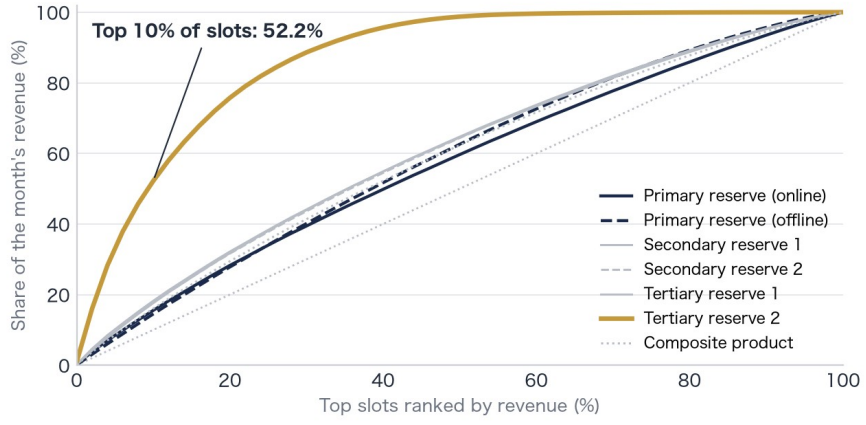


Figure 8 Share of balancing market sales concentrated in the top slots (August 2026 (1–31), nationwide)

Source: Trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

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In August 2026, there were 1,345 slots with winning bids. The top 1% of these slots by sales (average winning price × amount won) accounted for 8.4% of the month's sales. There were 34 slots with a nationwide average winning price of 10 yen/kW per 30 min or more, and 143 slots with no winning bids.

Slots with high maximum winning prices (by area where the power source is). The maximum winning price is the price of the single highest winning bid. The average winning price for the same slot and area is shown on the right:

Date	Time	Area	Maximum winning price (yen/kW per 30 min)	Average winning price for the same slot and area (yen/kW per 30 min)
August 6	9:00	Tokyo	97.00	51.63
August 6	9:30	Tokyo	97.00	68.96
August 6	10:00	Tokyo	97.00	45.53
August 6	11:00	Tokyo	97.00	35.21
August 9	9:00	Tokyo	97.00	12.73

Source: Tertiary reserve 2 trading results (preliminary figures) published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

10. ② Wholesale electricity (JEPX) | Average prices

Wholesale electricity is the electricity that power companies and retailers buy and sell. Its price is set in the "spot market" of the Japan Electric Power Exchange (JEPX), where electricity for the next day is traded the day before. "System (nationwide)" is the price when all of Japan is treated as one market (system price). When the transmission lines between regions are congested, prices split by area from Hokkaido to Kyushu (area prices).

The monthly average for System (nationwide) in wholesale electricity (JEPX) was 19.14 yen/kWh in August 2026, against 11.98 yen/kWh in the same month last year (a difference of 7.15 yen/kWh).

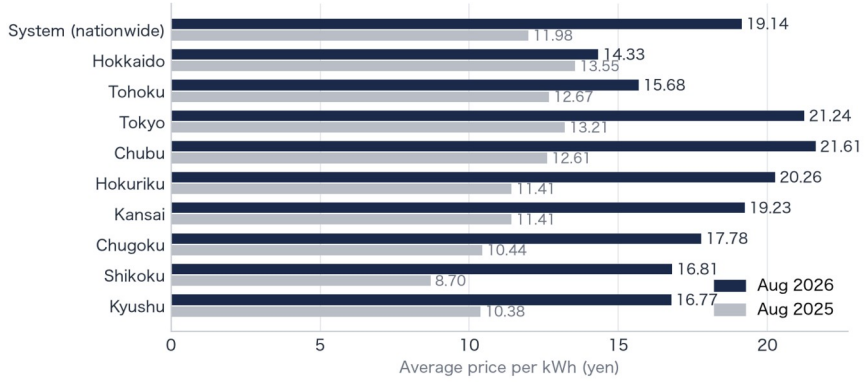


Figure 9 Average wholesale electricity price (yen/kWh). Navy is August 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of 30-minute prices. Compiled by us.

11. ② Wholesale electricity (JEPX) | How prices move during the day

Wholesale prices change every 30 minutes. The figure shows the shape of a typical day: the prices at the same time of day, averaged over the month. If a battery charges in cheap hours and discharges in expensive hours, the difference becomes its income.

The daily low and high for wholesale electricity were 17.32 yen/kWh around 6:30 and 28.06 yen/kWh around 17:30 in Tokyo. In Kyushu, they were 10.55 yen/kWh around 12:00 and 30.21 yen/kWh around 18:00.

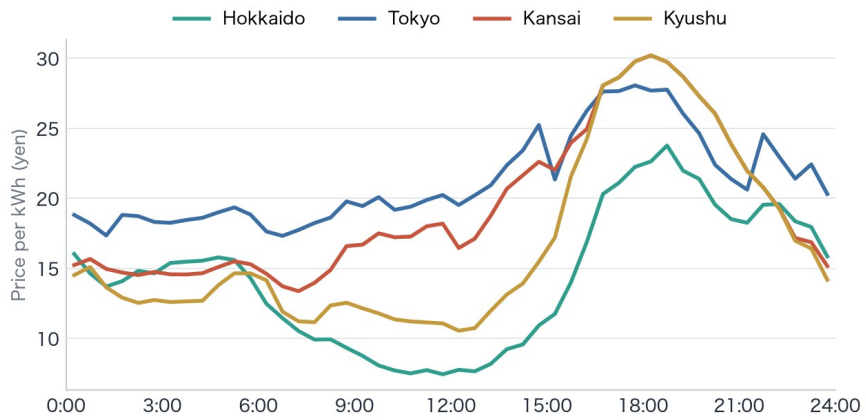


Figure 10 Daily price peaks and troughs for wholesale electricity (August 2026 average, yen/kWh)
Source: JEPX spot market. Compiled by us.

12. ② Wholesale electricity (JEPX) | How big is the price gap per kWh

For each day, we took the average price of the 4 most expensive hours and subtracted the average price of the 4 cheapest hours. The figure shows the monthly average of this gap by area. It equals the gain per kWh from buying 1 kWh in the 4 cheapest hours and selling it in the 4 most expensive hours. It is a simple difference. It does not subtract electricity lost in charging and discharging, or fees. It also ignores whether the cheap hours come before the expensive hours.

By area, the monthly average gap between the 4 most expensive hours and the 4 cheapest hours in wholesale electricity was largest in Shikoku at 19.67 yen/kWh, and smallest in Tokyo at 11.94 yen/kWh.

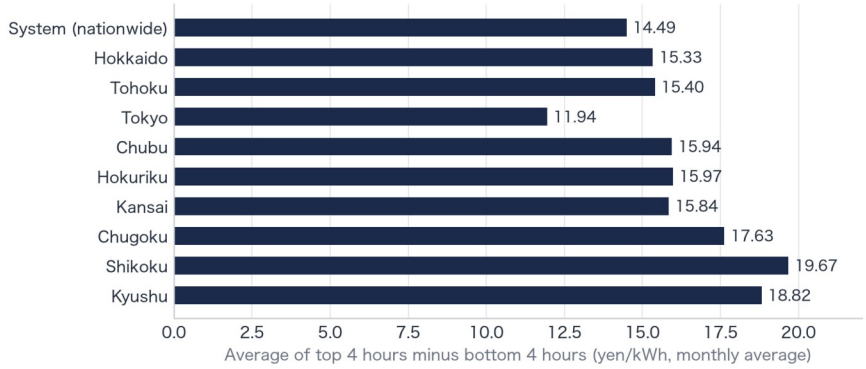


Figure 11 Wholesale price gap: average of the 4 most expensive hours − average of the 4 cheapest hours (August 2026, yen/kWh, monthly average)
Source: JEPX spot market. Compiled by us.

Area	2-hour gap (yen/kWh)	4-hour gap (yen/kWh)	Days with a 4-hour gap of 10 yen or more
System (nationwide)	16.09	14.49	26
Hokkaido	17.04	15.33	25
Tohoku	17.10	15.40	25
Tokyo	13.58	11.94	21
Chubu	17.46	15.94	29
Hokuriku	17.89	15.97	24
Kansai	17.56	15.84	23
Chugoku	19.62	17.63	24
Shikoku	22.29	19.67	27
Kyushu	20.53	18.82	24

The gaps are monthly averages of the daily gaps. August 2026 covers 31 days.

13. ② Wholesale electricity (JEPX) | There is a price gap every day

This shows the same 4-hour gap, day by day. The bigger the gap on a given day, the more room a battery has to earn from it.

Out of 31 days, the 4-hour gap in wholesale electricity was 10 yen/kWh or more on 26 days for System (nationwide), 29 days in Chubu and 21 days in Tokyo.

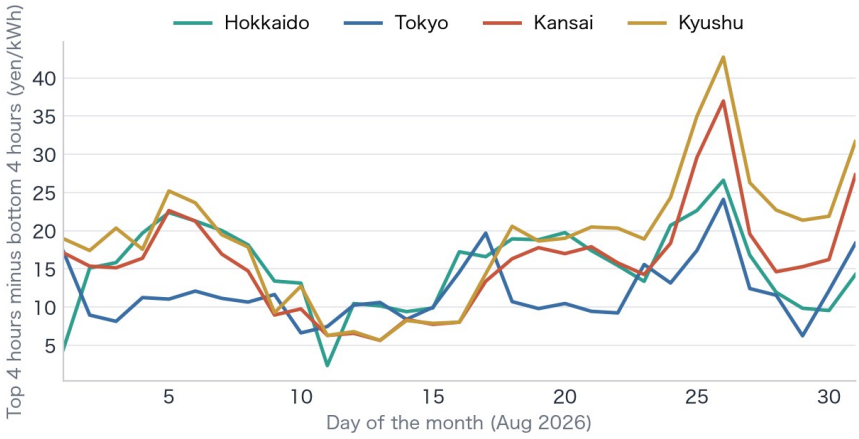


Figure 12 Daily wholesale price gap (average of the day's 4 most expensive hours − average of the 4 cheapest hours, yen/kWh)

Source: JEPX spot market. Compiled by us.

14. ③ Capacity market (OCCTO) | A market that buys future supply capacity

The capacity market buys electricity supply capacity for 4 years ahead, through an auction held once a year. Prices are set in yen per kW per year. The left chart shows the clearing prices of past main auctions. The right chart shows the battery and pumped hydro capacity that won in the Long-Term Decarbonization Power Source Auction. This is a separate auction that promises capacity income, in principle for 20 years, to things like new decarbonized power sources. Results are published once a year, so they do not change every month. The latest is for delivery FY2029 (published January 2026).

Clearing prices for Tokyo in the capacity market main auction were 14,137 yen/kW for delivery FY2024, 3,495 yen/kW for FY2025, 5,834 yen/kW for FY2026, 9,555 yen/kW for FY2027, 14,812 yen/kW for FY2028 and 15,111 yen/kW for FY2029. In the Long-Term Decarbonization Power Source Auction (bids in FY2025), the battery and pumped hydro capacity that won was 819 MW for lithium-ion batteries and pumped hydro (excluding new builds), and 886 MW for non-lithium-ion batteries, pumped hydro (new builds) and LDES. Both come from the latest once-a-year publication, so they do not change every month.

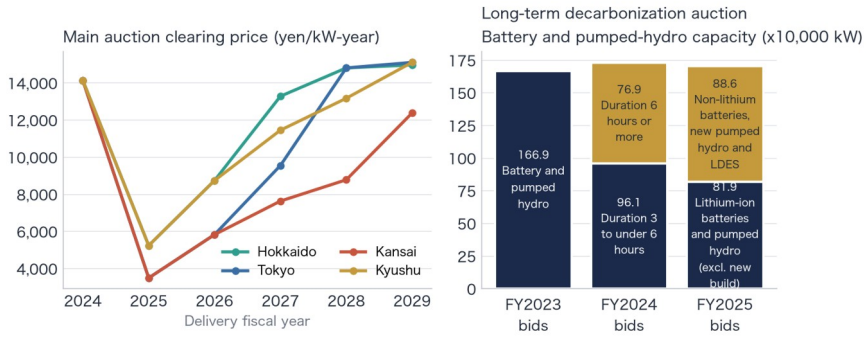


Figure 13 Capacity market clearing prices (left, yen/kW) and battery and pumped hydro capacity that won in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Clearing results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

15. Our records | Where battery storage projects are increasing

We counted the projects newly listed this month on our DealRoom (our data room for buyers, which lists battery storage projects), by the grid operator's grid area. These are figures for projects we handled, not counts for the whole market. Transfer of rights is a sale that hands over the land and power rights. Completed-build handover is a sale that hands over the plant once it is built.

New listings on our DealRoom in August 2026 were 17 projects, 175.6 MW. By grid area: Kyushu 6 projects, 105.6 MW; Chugoku 5 projects, 10.0 MW; Chubu 4 projects, 8.0 MW; Tokyo 1 project, 50.0 MW; Kansai 1 project, 2.0 MW.

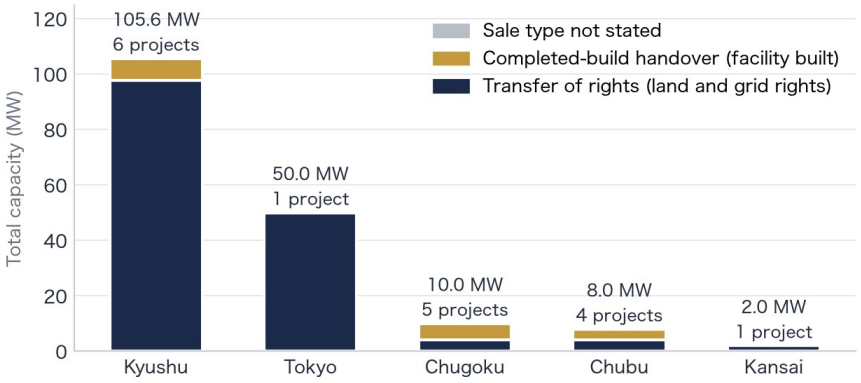


Figure 14 Output of projects we listed on our DealRoom in August 2026 (MW)

Source: Records of our DealRoom (project locations are given only down to the grid area; prices and parties are not listed).

16. Our records | How many months and how much it takes to connect to the grid

A grid connection study is a process where you ask the grid operator to check: "Can we connect at this site, and how much will the work cost?" The grid connection cost contribution is the part of the connection work cost that the applicant pays. From the response letters we received, we compiled only the high-voltage ones (connecting to 6,600 V distribution lines), by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

Median responses to grid connection studies (high voltage only, as of October 19, 2026) ranged from 8.0 months in Tokyo (n=48) to 17.0 months in Tohoku (n=5) for construction time. For the grid connection cost contribution, they ranged from ¥3.02 million in Tokyo (n=50) to ¥32.35 million in Tohoku (n=8).

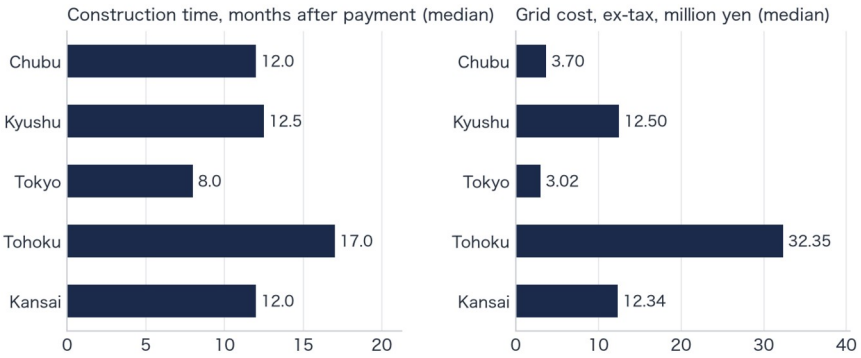


Figure 15 Median responses to grid connection studies (high voltage) by grid area (left: months of construction work, right: grid connection cost contribution, excluding tax)

Source: Compilation of the high-voltage grid connection study response letters we received from grid operators (as of October 19, 2026).

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17. Market figures (tables you can use as they are)

We put the figures behind the charts into tables that you can use in your own documents: ① nationwide figures by product for the balancing market (bid ratio, clearing rate, price, battery share), ② figures by area and trading volume for wholesale electricity (JEPX), and ③ clearing prices in the capacity market.

Balancing market (August 2026, preliminary figures, nationwide)

Product	Amount sought (MW)	Amount bid (MW)	Amount won (MW)	Bid ratio (bid ÷ sought)	Clearing rate (won ÷ bid)	Average winning price (yen/kW per 30 min)	Battery share
Primary reserve (online)	1,507.8	1,748.3	1,417.9	116.0%	81.1%	4.40	24.4%
Primary reserve (offline)	497.4	365.0	238.2	73.4%	65.3%	9.63	100.0%
Secondary reserve 1	1,180.9	1,838.8	1,243.4	155.7%	67.6%	3.01	1.6%
Secondary reserve 2	973.0	3,690.2	2,501.1	379.3%	67.8%	2.73	0.8%
Tertiary reserve 1	2,963.3	5,474.0	3,279.9	184.7%	59.9%	2.63	0.6%
Tertiary reserve 2	484.1	1,422.5	459.0	293.8%	32.3%	2.03	0.3%
Composite product	3,783.2	6,207.0	3,871.6	164.1%	62.4%	3.15	9.0%

Amounts are averages per 30-minute slot. A product with a bid ratio under 100% did not get enough bids for the amount sought. The clearing rate is the share of the amount bid that won.

Source: Trading results (preliminary figures) and trading results by power source type, published by the Electric Power Reserve Exchange (EPRX). Compiled by us.

Wholesale electricity, JEPX spot market (August 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour gap	Highest	30-minute slots at 30 yen or more	30-minute slots at 0.01 yen
System (nationwide)	19.14	11.98	16.72	27.60	14.49	50.01	127	0
Hokkaido	14.33	13.55	9.09	21.29	15.33	47.50	34	16
Tohoku	15.68	12.67	10.21	22.55	15.40	47.50	35	0
Tokyo	21.24	13.21	20.88	26.97	11.94	50.01	141	0
Chubu	21.61	12.61	20.00	30.10	15.94	55.23	253	0
Hokuriku	20.26	11.41	19.52	28.82	15.97	55.23	221	0
Kansai	19.23	11.41	18.72	28.42	15.84	55.23	179	0
Chugoku	17.78	10.44	14.41	28.39	17.63	55.23	145	0
Shikoku	16.81	8.70	12.12	27.65	19.67	55.23	141	10
Kyushu	16.77	10.38	13.01	28.33	18.82	55.23	143	0

Trading volume (August 2026, total for 31 days): total cleared volume 38,580 million kWh (same month last year: 26,819 million kWh), sell bids 58,644 million kWh, buy bids 48,003 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Compiled by us.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Battery and pumped hydro capacity that won in the Long-Term Decarbonization Power Source Auction (in units of 10,000 kW)

Bid year	Category	Capacity won (10,000 kW)
Bids in FY2023	Batteries and pumped hydro	166.9
Bids in FY2024	Continuous operation of 3 hours or more and under 6 hours	96.1

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Bid year	Category	Capacity won (10,000 kW)
Bids in FY2024	Continuous operation of 6 hours or more	76.9
Bids in FY2025	Lithium-ion batteries and pumped hydro (excluding new builds)	81.9
Bids in FY2025	Non-lithium-ion batteries, pumped hydro (new builds) and LDES	88.6

Source: Clearing results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

18. Assumptions and terms

- We compiled the figures from the values published by the sources. We have not added up several months or converted anything to annual figures. We do not show revenue amounts for any single battery storage plant.
- Months covered: for the balancing market (EPRX), preliminary figures for August 2026 (delivery dates up to the 31st); for wholesale electricity (JEPX), August 2026 (31 days). For the balancing market, the month may differ until the published values for August 2026 are included in our compilation.
- Balancing market: EPRX preliminary figures, nationwide (the total column). The amount sought is by grid operator (TSO). The amount bid, amount won and average winning price are by area where the power source is. Amounts are averages per 30-minute slot. Average winning prices are weighted by the amount won. Bid ratio = amount bid ÷ amount sought; clearing rate = amount won ÷ amount bid. For the 3-hour blocks up to March 13, 2026, we counted each block as 6 slots of 30 minutes.
- Range of winning prices: the thin line runs from the cheapest bid (lowest winning price) to the most expensive bid (highest winning price) among the month's slots with winning bids (published values by area where the power source is). The published values for Primary reserve (online), Secondary reserve 1 and 2, and Tertiary reserve 1 also include Composite product bids (one price for several products combined). Some bids are above the price cap of 15 yen (for deliveries up to August 31, 2026) for Primary reserve (online) and Secondary reserve 1, or above 10 yen for Secondary reserve 2 and Tertiary reserve 1. We removed such bids from the thin line when Composite product bids in the same slot and area can explain them. For Primary reserve (offline), we removed bids above the price cap (15 yen this month) from the thin line. (The balancing market system also clears bids above the cap, then cuts them down to the cap at settlement.) The thick band is the range from the bottom 10% to 90% of slots with winning bids, sorted by average winning price from cheapest. The dot is the average weighted by the amount won.
- Clearing rate by grid area × time of day: total amount won ÷ total amount bid, for slots at the same time of day across the month (by area where the power source is).
- The response time and duration of balancing market products are the FY2026 specifications (Electric Power Reserve Exchange, "Product requirements", 6th edition). Primary reserve senses frequency changes by itself and responds. Online is the category where the grid operator watches its operation over a dedicated line. Offline is the category where no dedicated line is laid, and the record of operation is submitted afterwards. For offline, the response time is within 30 seconds, and there is no set duration.
- High-priced slots for Tertiary reserve 2: slots with a nationwide average winning price of 10 yen/kW per 30 min or more. The top 1% means the top 1% of slots ranked by sales (average winning price × amount won), with 1% of the number of slots rounded to the nearest whole number. The maximum winning price is the published value by area where the power source is. It is the price of the single highest winning bid. In the balancing market, each winning bid is settled at its own bid price (pay-as-bid (multi-price)). So the average winning price for the same slot and area is lower than the maximum.
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime covers the slots from 8:00 to 16:00, and evening covers the slots from 16:00 to 20:00.
- Price gap: we sort the day's 30-minute slots by price. The gap is the average of the 8 most expensive slots (4 hours) minus the average of the 8 cheapest slots (4 slots for 2 hours). We do not consider the order of cheap and expensive hours, electricity lost in charging and discharging, or fees, taxes and transmission charges. It does not show actual revenue.

Sources (OCCTO, capacity market clearing results): delivery FY2024
https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025
https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026
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https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / delivery FY2028
https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028

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.pdf / delivery FY2029
https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): bids in FY2023 https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / bids in FY2024 https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / bids in FY2025 https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results for each delivery fiscal year were published in different years (FY2024 in 2020, FY2029 in January 2026). We do not show the number of battery projects in the Long-Term Decarbonization Power Source Auction, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary figures) and trading results by power source type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity securing contract documents / our compilation of the grid connection study response letters we received / records of our DealRoom.

Terms: system price (System (nationwide)) = the wholesale electricity price when all of Japan is treated as one market. Area price = the price that splits by area depending on transmission line congestion. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = a measure of power (1 MW = 1,000 kW). MWh = the amount of electricity that can be stored. yen/kWh = the price per 1 kWh of electricity.

Science X Inc. https://www.scix.co.jp Contact: s@scix.co.jp

Newsletter "Japan grid-scale BESS: primary data":
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